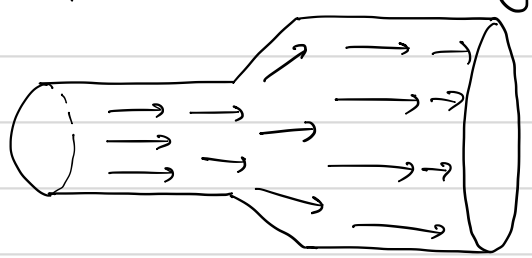
# Vector Fields (16.1)

functions

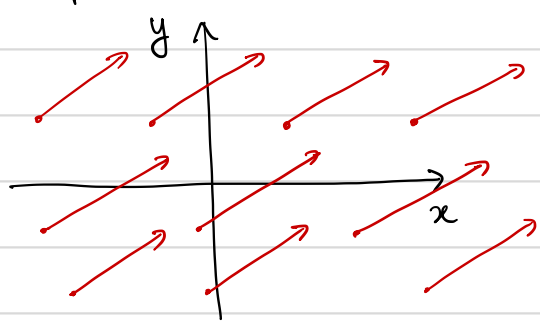
**2D:** $\vec{F} = \langle P(x,y), Q(x,y) \rangle$
$= P\hat{i} + Q\hat{j}$

**3D:** $\vec{F} = \langle P(x,y,z), Q(x,y,z), R(x,y,z) \rangle$
$= P\hat{i} + Q\hat{j} + R\hat{k}$

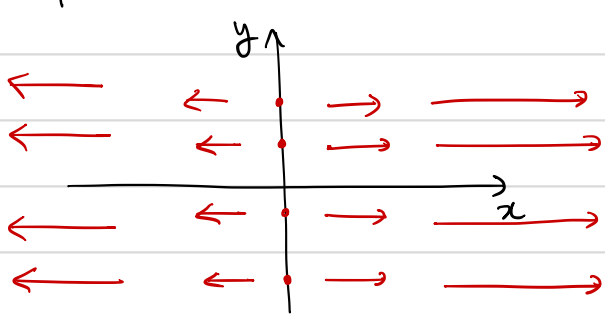
**Example:** Fluid velocity



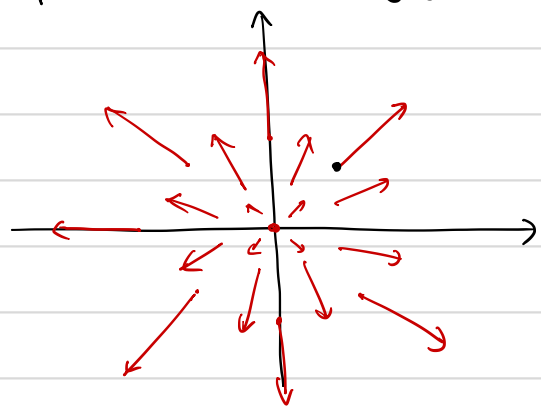
**Example:** $\vec{F} = 2\hat{i} + \hat{j}$



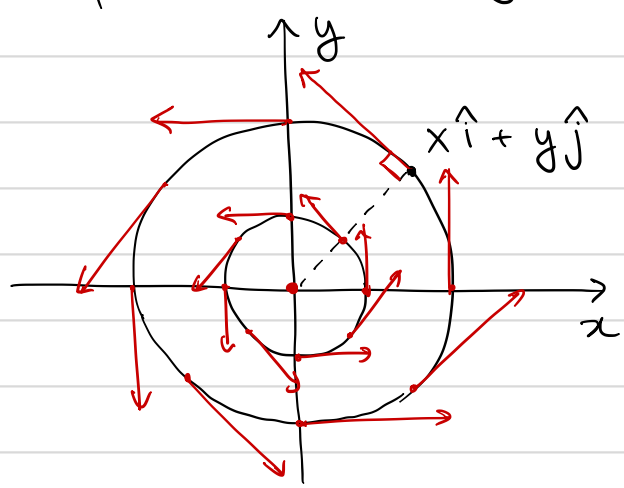

**Example:** $\vec{F} = x\hat{i}$




**Example:** $\vec{r} = x\hat{i} + y\hat{j}$

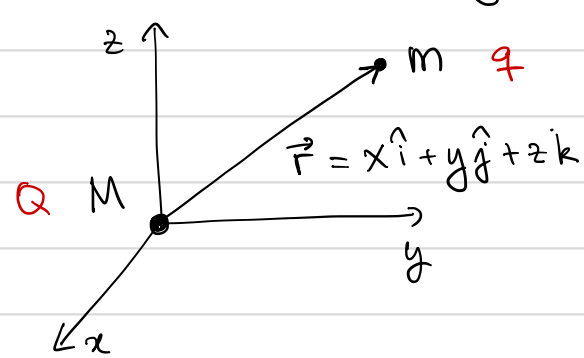


**Example:** $\vec{F} = -y\hat{i} + x\hat{j}$




Velocity field of uniform rotation w/ angular speed 1

**Example:** Newtonian gravity



Force of M on m:

$$|\vec{F}| = \frac{GMm}{|\vec{r}|^2}$$

$$\vec{F} = -\frac{GMm}{|\vec{r}|^3}\vec{r}$$

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<u>Example</u>: Electrostatics

$$\vec{F} = \frac{\varepsilon_0 Qq}{|\vec{r}|^3}\vec{r}$$

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<u>Example</u>: Gradient fields

$$\vec{F} = \nabla f = \langle f_x, f_y, f_z \rangle$$